Information:

Your homework has now been checked; no correction was needed except in **Question 2.4**. The homework is very related to what you have seen in the class Exercise today (specifically **2.1** *Fitting a non-linear regression by hand*).

```
def sinefun(params):
    a, b, c, log_s = params
    s = np.exp(log_s)
    yhat = a + b * np.sin(c * x)
    return np.sum(norm.logpdf(y, loc=yhat, scale=s))

# Example call
params_init = np.array([1, 1, 1, 1])
sinefun(params_init)
```

1. From the **van Bertalanffy growth function**: you can already notice that there are 3 parameters ($\mathbf{L_{inf}}$, **k**, and $\mathbf{t_0}$) and 1 variable (**x**). That is why you were given only 3 parameter values in **Question 1.2**. However, to plot this your defined function must return a function that includes **x** as seen in the growth function formula, so that this function will be defined for any given **x** values. For instance, is your growth function is called **GF**. Then for some: x-values = 34,23,12,13,53,22,33,45,34,23,19; the **GF(x)**, can be derived and plotted, provided that in **GF** the 3 parameters are already defined.

2. When you want to build an optimization function, the parameters of interest become 4 because of **s** as seen in the picture above. Remember the 4th given parameter is NOT **s** but **log_s**, so you have to exponentiate it to have **s** (again see picture above). By now you should know what is **y** and what is **x** for you. **Age** is **x** and **tl** is **y** (the variables you plotted in **Question 1.1**).

3. **yhat** is the GF(x) given any x and the 4 parameter values.

4. Finally, the only correction here is **Question 2.4**, it should be Plot fits (with CIs), because **2.5** asks about the intervals of **2.4**. Also you are only plotting 2 models instead of 3 (reason: the third model was skipped).